## Differentiation 1

**Pure Mathematics Year 2**

Chapter 9: Differentiation

## Chapter Overview 2

In Year 1 we differentiated terms of the form $ax^n$. This chapter extends those methods.

1. **Trigonometric, exponential and logarithmic functions**

Find $\frac{d}{dx}\sin x$, $\frac{d}{dx}\ln x$ and $\frac{d}{dx}2^x$.

2. **Chain, product and quotient rules**

Differentiate composite functions, products and quotients, such as $x\sin x$ and $e^{2x^2}$.

3. **Parametric differentiation**

If $x = \sin t$, $y = e^t$, find the tangent at $(0, 1)$.

4. **Implicit differentiation**

An implicit relationship need not give $y$ explicitly in terms of $x$.

If $xy + y^2 = 3$, find $\frac{dy}{dx}$.

5. **Rates of change**

A circle's radius increases at $5\ \text{cm/s}$. How fast does its area change when $r = 3\ \text{cm}$?

## Differentiating Trigonometric Functions 3

**Differentiate $\sin x$ from first principles.**

Recall, as $h \to 0$, $\sin h \approx h$, $\cos h \approx 1 - \frac{1}{2}h^2$.
Also $\sin(x+h) = \sin x \cos h + \cos x \sin h$.

**First principles and addition formula**

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$
$$= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}.$$

**Collect the sine and cosine terms**

$$\frac{dy}{dx} = \lim_{h\to 0} \left( \frac{\cos h - 1}{h} \sin x + \frac{\sin h}{h} \cos x \right).$$

**Use the small-angle limits**

$$\lim_{h\to 0} \frac{\sin h}{h} = 1, \qquad \lim_{h\to 0} \frac{\cos h - 1}{h} = 0.$$

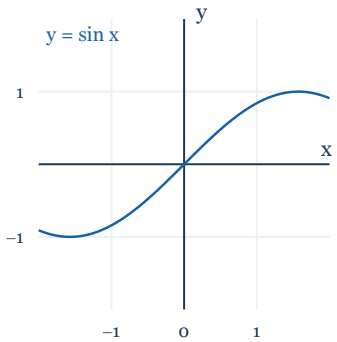
The approximations give $h/h = 1$ and $-h^2/(2h) = -h/2 \to 0$.

$$\frac{dy}{dx} = 0 \sin x + 1 \cos x = \cos x.$$

## Why Must We Use Radians? 4

**Explanation 1**

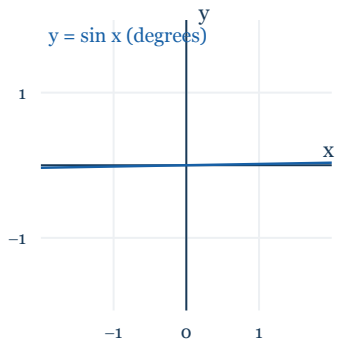
The approximations $\sin x \approx x$ and $\cos x \approx 1 - \frac{1}{2}x^2$, as $x \to 0$, require radians. The first-principles proof used these approximations.




In radians, $\cos 0 = 1$, so the predicted gradient at the origin is 1. This agrees with the graph.

**Explanation 2**

Compare the graphs with the same scale on both axes.




In degrees, the gradient at the origin is much smaller than 1. The formula $\frac{d}{dx}\sin x = \cos x$ therefore does not apply to degree inputs.

## Differentiating Trigonometric Functions 5

Find $\frac{d}{dx}\cos x$.

$$\frac{d}{dx}\cos x = -\sin x.$$

**Memorisation tip**

Moving right differentiates; moving left integrates.

$$\sin x \longrightarrow \cos x$$

For the "wrong" function, the sign changes: differentiating $\cos x$, or integrating $\sin x$.

## Differentiating Sine and Cosine of kx 6

Find $\frac{d}{dx}\sin kx$.

$k\cos kx$

Find $\frac{d}{dx}\cos kx$.

$-k\sin kx$

These results come from the chain rule, later in this chapter.

**Quickfire questions**

a. $\frac{d}{dx}\sin 3x$

a

d. $\frac{d}{dx}(4\cos 3x)$

d

b. $\frac{d}{dx}\cos 5x$

b

e. $\frac{d}{dx}(-\frac{1}{2}\sin x)$

e

c. $\frac{d}{dx}(3\sin 5x)$

c

f. $\frac{d}{dx}(-\frac{2}{3}\cos \frac{x}{2})$

f

## Example — Stationary Points 7

A curve has equation $y = \frac{1}{2}x - \cos 2x$. Find its stationary points for $0 \leq x \leq \pi$.

Working

## Test Your Understanding — Stationary Points 8

A curve has equation $y = \sin 3x + 2x$. Find its stationary points for $0 \leq x \leq \frac{2}{3}\pi$.

Working

## Exercise 9A — Page 234 9

1. a. Given $f(x) = \cos x$, show that

$$f\prime(x) = \lim_{h \to 0} \left( \frac{\cos h - 1}{h} \cos x - \frac{\sin h}{h} \sin x \right).$$

b. Hence prove that $f\prime(x) = -\sin x$.

1a

1b

## Exercise 9A — Page 234 10

7. A curve has equation $y = 2\sin 2x + \cos 2x$.
Find its stationary points for $0 \leq x \leq \pi$.

7

8. A curve has equation $y = \sin 5x + \cos 3x$. Find the equation of its tangent at $(\pi, -1)$. **(4 marks)**

8



9. A curve has equation $y = 2x^2 - \sin x$. Show that its normal at $x = \pi$ has equation $x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$. **(7 marks)**

9

10. Prove from first principles that the derivative of $\sin x$ is $\cos x$.

You may use the addition formula for $\sin(A+B)$, and that as $h \to 0$,

$$\frac{\sin h}{h} \to 1, \qquad \frac{\cos h - 1}{h} \to 0.$$

10

## Differentiating Exponentials and Logarithms 14

**Recall from Year 1**

$$\frac{d}{dx}e^x = e^x.$$

**Just seen**

$$\frac{d}{dx}\sin kx = k\cos kx.$$

Applying the same input-scaling idea $\longrightarrow$

$$\frac{d}{dx}e^{kx} = ke^{kx}.$$

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

$$\frac{d}{dx}a^x = a^x \ln a$$

$$\frac{d}{dx}a^{kx} = ka^{kx} \ln a$$

For positive $k, x$,

$$\frac{d}{dx}\ln(kx) = \frac{d}{dx}(\ln k + \ln x) = \frac{1}{x}.$$

Scaling the logarithm's input does not change the gradient. Proofs follow using implicit differentiation.

## Quickfire Questions — 1/2 15

1. $\frac{d}{dx}(3^x)$

1

2. $\frac{d}{dx}(x^3)$

2

3. $\frac{d}{dx}(\ln(3x))$

3

4. $\frac{d}{dx}(3^{2x})$

4

5. $\frac{d}{dx}(2^{3x})$

5

6. $\frac{d}{dx}(5\ln x)$

6

7. $\frac{d}{dx}(e^{x/2})$

7

8. $\frac{d}{dx}(5\ln(2x))$

8

9. $\frac{d}{dt}(9^t)$

9

10. $\frac{d}{dx}(5(4^x))$

10

11. $\frac{d}{dx}(x^4)$

11

12. $\frac{d}{dx}(\ln(6x))$

12

13. $\frac{d}{dx}(6\ln x)$

13

14. $\frac{d}{dx}(3e^{2x})$

14

15. $\frac{d}{dx}(e^{-x})$

15

## Quickfire Questions — 2/2 16

**Applied example**

$P = 1000(2^t)$ models a rabbit population after $t$ years. When does its rate of increase reach 20,000 rabbits per year?

Application

## Test Your Understanding 17

Differentiate $y = (e^x + 2)^2$.

**Hint: Expand first.**

Derivative

A child's headlice are treated with shampoo. The population after $t$ days is modelled by $P = 460(3^{-2t})$.

a. When are 20 headlice left?

b. Find the rate of change after 3 days.

a

b

## Exercise 9B — Pages 236–237 18

4. Find the tangent to $y = 2^x + 2^{-x}$ at $(2, \frac{17}{4})$.

4

5. A curve has equation $y = e^{2x} - \ln x$. Show that the tangent at $x = 1$ is $y = (2e^2 - 1)x - e^2 + 1$.
**(6 marks)**

5



7. Cambridge's population was 37,000 in 1900 and about 109,000 in 2000. Model the population $t$ years after 1900 by $P = P_0 k^t$.

a. Find $P_0$ and $k$.

b. Evaluate $dP/dt$ in 2000.

c. Interpret your answer.

7a

7b

7c

9. Prove that $\frac{d}{dx}a^{kx} = ka^{kx}\ln a$, assuming $\frac{d}{dx}e^{kx} = ke^{kx}$.

9

10. $f(x) = e^{2x} - \ln(x^2) + 4$, $x > 0$.

a. Find $f'(x)$.

b. The curve has gradient 2 at $x = a$. Show that $a(e^{2a} - 1) = 1$.

10a

10b



11. Curve $C$ has equation $y = 5\sin 3x + 2\cos 3x$, $-\pi \le x \le \pi$.

a. Show that $P(0, 2)$ lies on $C$.

b. Find the normal at $P$.

11a

11b

12. Point $P$ lies on $y = 2(3^{4x})$ with $x = 1$. Find the normal at $P$ in the form $y = ax + b$, giving $a, b$ exactly.

12

## For Your Interest — Differentiating ln x 24

Prove from first principles that $\frac{d}{dx}\ln x = \frac{1}{x}$, for $x > 0$.

**First principles; then the logarithm law**

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{\ln(x+h) - \ln x}{h}$$
$$= \lim_{h\to 0}\frac{\ln((x+h)/x)}{h}$$
$$= \lim_{h\to 0}\frac{1}{h}\ln\left(1+\frac{h}{x}\right).$$

**Move the multiplier into the power**

$$= \lim_{h\to 0}\ln\left(1+\frac{h}{x}\right)^{1/h}.$$

**Substitute $n = 1/h$**

For the right-hand limit, $n \to \infty$:

$$\lim_{n\to\infty}\ln\left(1+\frac{1/n}{x}\right)^{n}$$
$$= \ln\left(\lim_{n\to\infty}\left(1+\frac{1/x}{n}\right)^{n}\right).$$

Here $(1/n)/x = (1/x)/n$. Continuity of $\ln$ lets us move the limit inside.

Using $e^{k} = \lim_{n\to\infty}(1+k/n)^{n}$, the result is $\ln(e^{1/x}) = 1/x$. The left-hand limit agrees by the same exponential limit.

Hence $\frac{dy}{dx} = \frac{1}{x}$.

## Differentiating Combinations of Functions 25

Functions can interact in different ways. Choose the matching rule.

1. **Composite function** $\longrightarrow$ **Chain rule**

$y = f(g(x)) = \sqrt{1 + 3x}$.
Outer function: $f(x) = \sqrt{x}$.
Inner function: $g(x) = 1 + 3x$.

Exercise 9C

2. **Product of two functions** $\longrightarrow$ **Product rule**

$y = f(x)g(x) = x \sin 2x$.

Exercise 9D

3. **Quotient of two functions** $\longrightarrow$ **Quotient rule**

$y = f(x)/g(x) = \ln x/x$.

Exercise 9E

## The Chain Rule 26

The chain rule differentiates a composite function: a function within a function.

Example: $y = (3x^4 + x)^5$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

**Note**

The $du$ symbols appear to cancel. This is a memory aid, not a proof.

**Full method**

Let $u = 3x^4 + x$: this is the inner function.

$$\frac{du}{dx} = 12x^3 + 1.$$

The outer function is $y = u^5$, so

$$\frac{dy}{du} = 5u^4 = 5(3x^4 + x)^4.$$

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

**Mentally in one go**

Differentiate the outer function, keeping its input unchanged, then multiply by the derivative of the inner function.

Think “something to the fifth power”: its outer derivative is five times that same “something” to the fourth power.

Outer derivative: $5(3x^4 + x)^4$.

Now differentiate the inner “something”, $3x^4 + x$.

Inner derivative: $12x^3 + 1$.

Multiply:

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

## Practice — Chain Rule — 1/2



| Original function | Outer derivative | Inner derivative |
|---|---|---|
| 1. $y = (x^2 + 1)^3$ | 1 outer<br>Differentiate the cube. | 1 inner<br>Multiply by the derivative of the inside function. |
| 2. $y = (\ln x)^3$ | 2 outer<br>Differentiate the cube. | 2 inner<br>Multiply by the derivative of the inside function. |
| 3. $y = e^{x^2+x}$ | 3 outer<br>Differentiate the exponential. | 3 inner<br>Multiply by the derivative of the exponent. |

## Practice — Chain Rule — 2/2



| Original function | Outer derivative | Inner derivative |
|---|---|---|
| 4. $y = (2^x + 1)^2$ | 4 outer<br>Differentiate the square. | 4 inner<br>Multiply by the derivative of the inside function. |
| 5. $y = \ln(\sin x)$ | 5 outer<br>Differentiate the natural logarithm. | 5 inner<br>Multiply by the derivative of the inside function. |

## Further Practice — Rewrite Before Differentiating — 1/2



1. $y = \sin 5x$

1 outer

Differentiate the sine.

1 inner

Differentiate the inside function; multiply the factors.

2. $y = \sin^2 x$

2 rewrite

Rewrite using brackets first.

2 outer

Differentiate the square.

2 inner

Differentiate the inside function; multiply the factors.

3. $y = \sqrt{x + 1}$

3 rewrite

Rewrite using brackets first.

3 outer

Differentiate the power of one half.

3 inner

Differentiate the inside function; multiply the factors.

## Further Practice — Rewrite Before Differentiating — 2/2



4. $y = \cos^3 2x$

4 rewrite

Rewrite using brackets first.

4 outer

Differentiate the cube.

4 inner

Use the chain rule twice.

5. $y = e^{e^x}$

5 outer

Differentiate the exponential.

5 inner

Differentiate the inside function; multiply the factors.

## Test Your Understanding — Chain Rule



**C3 June 2011, Q1a (2 marks)**

Differentiate $\ln(x^2 + 3x + 5)$.

Exam

**Textbook example**

Given $y = \sqrt{5x^2 + 1}$, find $dy/dx$ at $(4, 9)$.

Textbook

**In general**

$$\frac{d}{dx}\ln(f(x)) = \frac{f'(x)}{f(x)}.$$

## Finding dy/dx from dx/dy



Sometimes $x$ is given in terms of $y$, but we want $dy/dx$.

$$\frac{dy}{dx} = \frac{1}{dx/dy}$$

**Proof using the chain rule**

$$\frac{dy}{dy} = \frac{dy}{dx}\frac{dx}{dy}.$$

Since $dy/dy = 1$, divide by $dx/dy \neq 0$.

**Example 1**

Find $dy/dx$ when $x = 2y^2 + y$.

1

**Example 2**

Find the gradient of $x = (1 + 2y)^3$ when $y = 1$.

2

4. Curve $C$ has equation $y = (5 - 2x)^3$. Find its tangent at the point $P$ where $x = 1$.

4



5. Given $y = (1 + \ln 4x)^{3/2}$, find $dy/dx$ at $x = \frac{1}{4}e^3$.

5

9. a. Differentiate $e^y = x$ with respect to $y$.

b. Hence prove that $y = \ln x$ gives $dy/dx = 1/x$.

9a

9b

10. Curve $C$ has equation $x = 4\cos 2y$.

a. Show that $Q(2, \pi/6)$ lies on $C$.

b. Show that $dy/dx = -1/(4\sqrt{3})$ at $Q$.

c. Find the normal at $Q$, in the form $ax + by + c = 0$, with exact constants.

10a

10b

10c



12. Curve $C$ has equation $y = 4/(2 - 4x)^2$, $x \neq 1/2$. Point $A$ has $x = 3$. Find the normal at $A$ in the form $ax + by + c = 0$, with integer constants.

12

## Extension — STEP I 2014, Q4 37

[STEP I 2014 Q4] An accurate clock has an hour hand of length $a$ and a minute hand of length $b$ (where $b > a$), both measured from the pivot at the centre of the clock face. Let $x$ be the distance between the ends of the hands when the angle between the hands is $\theta$, where $0 \leq \theta < \pi$.

Show that the rate of increase of $x$ is greatest when $x = \left(b^2 - a^2\right)^{\frac{1}{2}}$.

In the case when $b = 2a$ and the clock starts at mid-day (with both hands pointing vertically upwards), show that this occurs for the first time a little less than 11 minutes later.

Setup

Maximum

Time

## The Product Rule 38

Use the product rule for a product of two functions:

$$y = uv \implies \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}.$$

### Method tips

Write out $u$, $v$, $u\prime$, $v\prime$: the individual derivatives can be complicated. Factorising is typical when finding turning points. Exponentials cannot equal zero.

### Memorisation tip

1. Differentiate one factor; leave the other.
2. Do it the other way round.
3. Add. Either order gives the same result.

### Example 1

Find $dy/dx$ for $y = x^2 \sin x$.

Write $u, v$, then differentiate each.

1 factors

Substitute into the product rule.

1 result

### Example 2

Find the turning point of $y = xe^{2x}$.

2 derivative

2 x

2 y

**Example 3**

If $y = e^{4x}\sin^2 3x$, show that

$$\frac{dy}{dx} = e^{4x}\sin 3x(A\cos 3x + B\sin 3x),$$

finding $A, B$.

3

**Example 4**

Find $f\prime(x)$ for $f(x) = x^2\sqrt{3x-1}$.

4

**Factoring tip**

Take out the smallest power, $-1/2$. The other power is one higher, leaving $(3x-1)$. Also take out $1/2$, common to $3/2$ and $2$.

The final radical form is optional, but avoids negative powers.

## Test Your Understanding — Product Rule 40

**Exam-style question 1 (4 marks)**

Differentiate $x^2\ln(3x)$, simplifying your answer.

1

**Question 2**

Given $\frac{dy}{dx} = \frac{1}{6x(x-1)^{1/2}}$, find $d^2y/dx^2$, simplifying your answer.

2

**Examiner comment**

“Perhaps the most challenging question in the paper.”

6. Curve $C$: $y = x^2\cos(x^2)$. Find the tangent at $P(\sqrt{\pi}/2, \pi\sqrt{2}/8)$ in the form $ax + by + c = 0$, with exact constants. **(7 marks)**

6 derivative

6 tangent



7. Given $y = 3x^2(5x - 3)^3$, show that $\dfrac{dy}{dx} = Ax(5x - 3)^n(Bx + C)$, finding $A, n, B, C$.

7

8. Curve $C$: $y = (x+3)^2 e^{3x}$.

a. Find $dy/dx$ using the product rule.

b. Find the gradient at $x = 2$.

8a

8b



9. a. Differentiate $(2\sin x - 3\cos x)\ln 3x$.

9a

9. b. Differentiate $x^4 e^{7x-3}$.

9b

10. Find $dy/dx$ at $x = 1$ on $y = x^5\sqrt{10x+6}$.

10

## The Quotient Rule 46

Use the quotient rule to differentiate a quotient, just as the product rule differentiates a product. Begin with $y = u/v$.

$$y = uv^{-1}$$

$$\frac{dy}{dx} = -uv^{-2}\frac{dv}{dx} + \frac{du}{dx}v^{-1}$$

$$= v^{-2}\left(v\frac{du}{dx} - u\frac{dv}{dx}\right)$$

$$= \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

**The quotient rule**

$$y = \frac{u}{v} \implies \frac{dy}{dx} = \frac{vu\prime - uv\prime}{v^2}.$$

**Memorisation tip: "Bottoms first!"**

The old denominator $v$ comes first in the numerator; square it in the new denominator. Note the subtraction in the numerator, unlike the addition in the product rule.

## Examples — Quotient Rule

47

**Example 1**

Find $dy/dx$ if $y = x/(2x + 5)$.

Set out the factors as for the product rule.

1

**Example 2**

Find the stationary point of $y = \sin x/e^{2x}$, $0 < x < \pi$.

2

**Hints for Example 2**

When a fraction equals zero, multiply by its nonzero denominator.

Factorise when one side is zero.

A sine/cosine mixture can be converted to tangent by dividing by cosine, checking it is nonzero.

## Test Your Understanding — 1/2

48

**Exam-style question 1 (5 marks)**

Differentiate $\sin 4x/x^3$, giving your answer in simplest form.

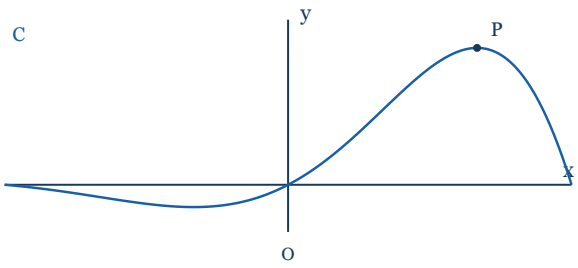
1

**Question 2**

Curve $C$: $y = e^{x\sqrt{3}} \sin 3x$, $-\pi/3 \leq x \leq \pi/3$.

a. Find the positive $x$-coordinate of turning point $P$, as a multiple of $\pi$. **(6 marks)**

b. Find the normal at $x = 0$. **(3 marks)**




2a

2b



6. Find the tangent to $y = e^{x/3}/x$ at $(3, e/3)$.

6

8. Curve $C$: $x = e^{y}/(3 + 2y)$.

a. Find its intercept $P$ on the $x$-axis. **(1 mark)**

b. Find the normal at $P$ in the form $y = mx + c$, with exact constants. **(6 marks)**

8a

8b



9. Differentiate $x^{4}/\cos 3x$ with respect to $x$.

9

10. Curve $C$: $y = e^{2x}/(x-2)^2$, $x \neq 2$.

a. Show that $dy/dx = Ae^{2x}(Bx - C)/(x-2)^3$, finding integers $A, B, C$.

b. Find the tangent at $x = 1$.

10a

10b



11. $f(x) = \frac{2x}{x+5} + \frac{6x}{x^2+7x+10}$, $x > 0$.

a. Show that $f(x) = 2x/(x+2)$.

b. Hence find $f\prime(3)$.

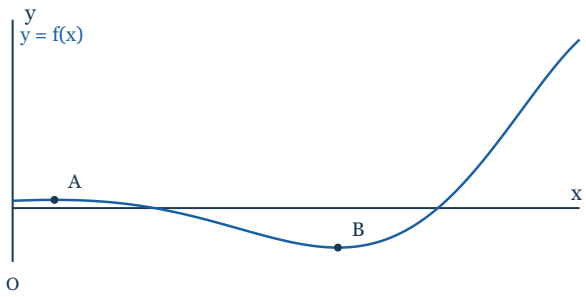
11a

11b

12. $f(x) = 2\cos 2x / e^{2-x}$, $0 < x < \pi$. The curve has a local maximum at $A$ and a local minimum at $B$.

a. Show that the $x$-coordinates of $A$, $B$ satisfy $\tan 2x = 1/2$. **(4 marks)**

b. Find the range of $f(x)$. **(2 marks)**




12a

12b

## Extension — Stationary Points 56

**STEP I 2010, Q2**

The curve $y = \frac{x-a}{x-b}e^x$, with constants $a, b$, has two stationary points. Show that $a - b < 0$ or $a - b > 4$.

Differentiate

Discriminant

## Differentiating Other Trigonometric Functions 57

Differentiate $y = \tan x$.

Use the definition of tangent, then the quotient rule.

$$y = \frac{\sin x}{\cos x}, \quad u = \sin x, \quad v = \cos x.$$
$$u\prime = \cos x, \quad v\prime = -\sin x.$$
$$\frac{dy}{dx} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

**More generally**

$$\frac{d}{dx}\tan kx = k\sec^2 kx.$$

**Tip**

This is in the formula booklet, but memorising it saves exam time.

Differentiate $y = \sec x$.

$$y = \frac{1}{\cos x} = (\cos x)^{-1}.$$
$$\frac{dy}{dx} = -(\cos x)^{-2}(-\sin x)$$
$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x}\frac{\sin x}{\cos x} = \sec x \tan x.$$

**Tip**

The quotient rule would work too; the chain rule is often simpler when the numerator is constant.

More generally,

$$\frac{d}{dx}\sec kx = k\sec kx \tan kx.$$

## Further Trigonometric Derivatives 58

$$\frac{d}{dx}\tan x = \sec^2 x.$$
$$\frac{d}{dx}\sec x = \sec x \tan x.$$
$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x.$$
$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x.$$

**Memorisation tip**

Adding “co” to the left-hand function adds “co” to the factors in its derivative, and changes the sign.

**Example 1**

Differentiate $y = \operatorname{cosec} 2x/x^2$.

1

**Example 2**

Differentiate $y = \sec^3 x$.

2

**Observation**

The power of secant stays the same when differentiating. This will matter in integration.

**Edexcel C3 June 2013(R), Q5b (3 marks)**

Show that $\frac{d}{dx}(\sec^2 3x)$ can be written as $u(\tan 3x + \tan^3 3x)$, where $u$ is constant.

Solution

## Using the Reciprocal Derivative 60

When $x$ is given in terms of $y$, we may want $dy/dx$ in terms of $x$. Choose an appropriate trigonometric identity:

$\sin^2 x + \cos^2 x \equiv 1, \qquad 1 + \tan^2 x \equiv \sec^2 x.$

**Example**

Given $x = \tan y$, express $dy/dx$ in terms of $x$.

Working

**Choosing the useful identity**

Writing $dy/dx = \cos^2 y$ is correct, but it does not yet express the answer in terms of the known $\tan y = x$.

The identity $1 + \tan^2 y = \sec^2 y$ makes that substitution directly.

## Further Examples — Inverse Trigonometric Functions 61

Show that $y = \arcsin x$ gives $dy/dx = 1/\sqrt{1-x^2}$.

Use $\sin^2 y + \cos^2 y = 1$ to express $\cos y$ in terms of $x = \sin y$.

$$y = \arcsin x \implies x = \sin y.$$

$$\frac{dx}{dy} = \cos y, \qquad \frac{dy}{dx} = \frac{1}{\cos y}.$$

For $-1 < x < 1$, the principal arcsine gives $-\pi/2 < y < \pi/2$, so $\cos y > 0$.

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$

Given $y = \arcsin(x^2)$, find $dy/dx$.

Example

**Formula booklet**

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}.$$

## Test Your Understanding — Reciprocal and Chain Rules 62

**Edexcel C3 January 2011, Q8b,c**

Given $x = \sec 2y$:

b. Find $dx/dy$ in terms of $y$. **(2 marks)**

c. Hence find $dy/dx$ in terms of $x$. **(4 marks)**

b

c

**Textbook**

Given $y = \arctan \frac{1-x}{1+x}$, find $dy/dx$. Use the full chain-rule method.

Textbook

5. Curve $C$: $y = 1/(\cos x \sin x)$, for $0 < x < \pi$ where defined.

a. Find $dy/dx$.

b. Determine the number of stationary points.

5a

5b



6. Show that $y = \sec x$ gives $dy/dx = \sec x \tan x$.

7. Show that $y = \cot x$ gives $dy/dx = -\operatorname{cosec}^2 x$.

6

7

**STEP I 2011, Q3**

Prove the identity

$4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta. \quad (*)$

i. By differentiating (*) or otherwise, show that

$$\cot\frac{\pi}{9}-\cot\frac{2\pi}{9}+\cot\frac{4\pi}{9}=\sqrt{3}.$$

ii. Set $\theta=\pi/6-\phi$ in (*), or use another method, to obtain a similar identity for $\cos 3\theta$. Deduce that

$$\cot\theta\cot(\pi/3-\theta)\cot(\pi/3+\theta)=\cot 3\theta.$$

Show that

$$\operatorname{cosec}\frac{\pi}{9}-\operatorname{cosec}\frac{5\pi}{9}+\operatorname{cosec}\frac{7\pi}{9}=2\sqrt{3}.$$



Prove

$4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta.$

Use two factor formulae:

$2\sin A\sin B=\cos(A-B)-\cos(A+B),$

$2\sin A\cos B=\sin(A+B)+\sin(A-B).$

These follow by expanding the right-hand sides using addition formulae and collecting terms.

Proof

**i. Differentiate the identity**

A product of several factors is differentiated one factor at a time, keeping the others, then adding.

$(uvwt\cdots)\prime = u\prime vwt\cdots + uv\prime wt\cdots$
$+uvw\prime t\cdots + \cdots .$

i



ii

Final identity

## Parametric Differentiation 69

Parametric equations define $x, y$ (and possibly $z$) in terms of another parameter, such as $t$.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad dx/dt \neq 0.$$

**Proof**

By the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

Rearrange to obtain the formula.

**Textbook example**

Find the gradient when $t = 2$ for $x = t^3 + t$, $y = t^2 + 1, t \in \mathbb{R}$.

Gradient

Find the normal at $\theta = \pi/6$ on $x = 3\sin\theta$, $y = 5\cos\theta, 0 \leq \theta < 2\pi$.

Normal

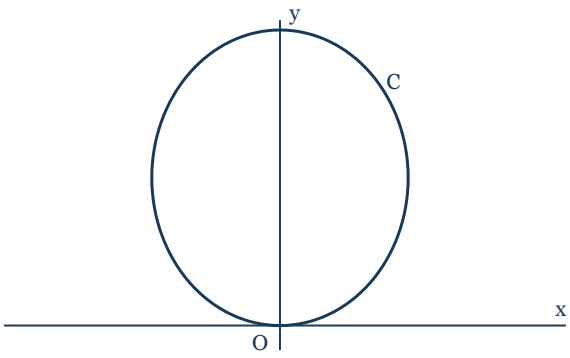
## Test Your Understanding — Parametric Tangent 70

**C4 June 2012, Q6**

Curve $C$: $x = \sqrt{3}\sin 2t$, $y = 4\cos^2 t$, $0 \leq t \leq \pi$.

a. Show that $dy/dx = k\sqrt{3}\tan 2t$, finding $k$. **(5 marks)**

b. Find the tangent at $t = \pi/3$, in the form $y = ax + b$. **(4 marks)**




a

b

2. a. Find the tangent to $x = 3 - 2\sin t$, $y = t\cos t$, at $t = \pi$.

2a

2. b. Find the tangent to $x = 9 - t^2$, $y = t^2 + 6t$, at $t = 2$.

2b



3. a. Find the normal to $x = e^t$, $y = e^t + e^{-t}$, at $t = 0$.

3a

3. b. Find the normal to $x = 1 - \cos 2t$, $y = \sin 2t$, at $t = \pi/6$.

3b

4. Find the points of zero gradient on

$$x = \frac{t}{1-t}, \qquad y = \frac{t^2}{1-t}, \quad t \neq 1.$$

You need not classify them as maxima or minima.

4



5. Curve $C$: $x = e^{2t}, y = e^t - 1, t \in \mathbb{R}$.

a. Find the tangent at $t = \ln 2$.

b. Show that $C$ has no stationary points.

5a

5b

6. Curve $C$: $x = (t^2 - 3t - 4)/t$, $y = 2t$, $t > 0$. A tangent $l_1$ is parallel to $y = x + 5$. Find $l_1$.

6



7. $x = 2\sin^2 t$, $y = 2\cot t$, $0 < t < \pi/2$.

a. Find $dy/dx$ in terms of $t$.

b. Find the tangent at $t = \pi/6$.

7a

7b

8. Curve $C$: $x = 4\sin t$, $y = 2\operatorname{cosec} 2t$, $0 < t < \pi$, where defined. Point $A = (2\sqrt{3}, 4\sqrt{3}/3)$.

a. Find $t$ at $A$.

b. Show that the normal $l$ at $A$ is $9x + 12y - 34\sqrt{3} = 0$.

8a

8b



9. Curve $C$: $x = t^2 + t$, $y = t^2 - 10t + 5$, $t \in \mathbb{R}$. At $P$ the gradient is 2.

a. Find $P$.

b. Find the tangent at $P$.

c. Show that this tangent does not meet $C$ again.

9a

9b

9c

10. Curve $C$: $x = 2\sin t$, $y = \sqrt{2}\cos 2t$, $0 < t < \pi$.

a. Find $dy/dx$ in terms of $t$.

b. Find the normal $l$ at $A$, where $t = \pi/3$, in the form $ax + by + c = 0$.

c. Prove that $l$ meets the curve only at $A$.

10a

10b

10c



11. $x = \cos t$, $y = \frac{1}{2}\sin 2t$, $0 \le t < 2\pi$.

a. Find $dy/dx$.

b. Find the tangent at $A$, where $t = \pi/6$.

c. Find the two other tangents $l_1, l_2$ parallel to the tangent at $A$.

11a

11b

11c

## Implicit Differentiation 81

An explicit equation makes $y$ the subject, for example $y = x^2 + 3x$, so we can calculate $y$ directly from $x$.

An implicit relation may instead be

$$x^2 + y^2 = 8x \quad \text{or} \quad \cos(x + y) = \sin y.$$

Even with $y = x^2$, we differentiate *both* sides with respect to $x$:

$$\frac{d}{dx}(y) = \frac{d}{dx}(x^2) \quad \Longrightarrow \quad \frac{dy}{dx} = 2x.$$

The derivative of $y$ with respect to $x$ is, by definition, $dy/dx$.

**Two things to remember**

1. Differentiate each side of the equation, using the chain rule where needed.
2. Differentiating $y$ with respect to $x$ gives $dy/dx$.

## Examples — Implicit Differentiation 82

**Use the chain rule**

"Something squared" differentiates to twice that "something", multiplied by its derivative.

For $y^2$, multiply $2y$ by $dy/dx$.

1. $\frac{d}{dx}(y^2)$

1

2. $\frac{d}{dx}(\sin y)$

2

3. $\frac{d}{dx}(e^y)$

3

4. $\frac{d}{dx}(xy)$

4

**In general**

$$\frac{d}{dx} f(y) = f\prime(y)\frac{dy}{dx}.$$

5. $\frac{d}{dx}(e^{x^2 y})$

5

6. $\frac{d}{dx}(\tan(x + y))$

6

7. $\frac{d}{dx}(x^2 + \cos y)$

7

**Why not just $y$ for the derivative of $xy$?**

Unlike the constant $a$ in $ax$, $y$ varies with $x$, so the product rule is needed. This is a very common exam calculation.

**Textbook**

Find $dy/dx$ in terms of $x, y$ where $x^3 + x + y^3 + 3y = 6$.

Differentiate both sides: remember that 6 becomes 0. Then make $dy/dx$ the subject.

1

Find $dy/dx$ at $(1, 1)$ where $e^{2x} \ln y = x + y - 2$.

2

**Tip**

Substitute the point early; there is no need to isolate $dy/dx$ first.

Unlike the Year 1 examples, the gradient here can depend on both $x$ and $y$.



**C4 January 2008, Q5**

Curve: $x^3 - 4y^2 = 12xy$.

a. Find the two points where $x = -8$. **(3 marks)**

b. Find the gradient at each. **(6 marks)**

2008 a

2008 b

**C4 June 2014(R), Q3**

$x^2 + y^2 + 10x + 2y - 4xy = 10.$

a. Find $dy/dx$, fully simplified. **(5 marks)**

b. Find $y$ when $dy/dx = 0$. **(5 marks)**

**Hint:** solve simultaneously with the original equation.

2014 a

2014 b

**Exam note**

This type of implicit-differentiation question is common.

## Exercise 9H — Pages 255–256 86

1. Write $u = y^n$ and use the chain rule to show that
$$\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}.$$

1

2. Use the product rule to show that
$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y.$$

2

7. Curve $C$ is described by
$2x^2 + 3y^2 - x + 6xy - 1 = 0$.
Find its tangent at $(1, -2)$ in the form $ax + by + c = 0$, with integer constants.

7



8. Curve $C$ has equation $3^x = y - 2xy$. Find the exact gradient at $(2, -3)$.

8

9. Find the exact gradient of
$\ln(y^2) = \frac{1}{2}x\ln(x-1), \quad x > 1,\ y > 0,$
at the point where $x = 4$.

9



10. Curve $C$ satisfies $\sin x + \cos y = 0.5$, with $-\pi < x < \pi$, $-\pi < y < \pi$.

a. Find $dy/dx$.

b. Find its stationary points.

10a

10b

11. Curve $C$: $ye^{-3x} - 3x = y^2$.

a. Find $dy/dx$ in terms of $x, y$.

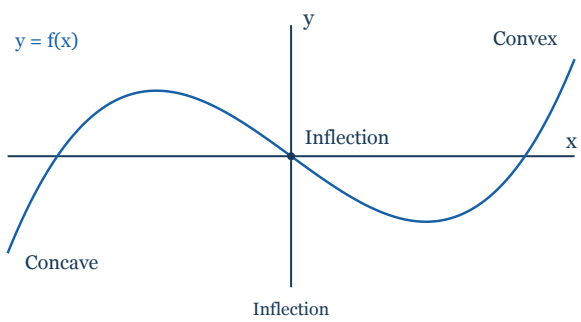
b. Show that the tangent at the origin is $y = 3x$.

11a

11b

## Using the Second Derivative 92

A point of inflection is where concavity changes: concave to convex, or vice versa. Informally, the curve changes the direction in which it is "swerving".




**Memorisation tip**

Picture a positive cubic, rising overall. Read "concave" then "convex" from left to right: alphabetical order.

**Concave**

The gradient decreases: $f''(x) \leq 0$ on the interval.

**Convex**

The gradient increases: $f''(x) \geq 0$ on the interval. The gradient may still be negative while increasing.

**Inflection**

When $f''$ is continuous, look for $f''(x) = 0$, then check that its sign changes. The gradient's rate of change is zero there, but the gradient itself need not be zero: an inflection need not be stationary.

## Examples — Concavity and Inflection

93

1. Find where $f(x) = x^3 + 4x + 3$ is concave.

1

2. Show that $f(x) = e^{2x} + x^2$ is convex for all real $x$.

2

3. Find the point of inflection of $y = x^3 - 2x^2 - 4x + 5$.

3

## Exercise 9I — Pages 259–261

94

9. Given $y = x(3x - 1)^5$:

a. Find $dy/dx$ and $d^2y/dx^2$.

b. Find the points of inflection.

9a

9b

11. A curve has equation $y = \frac{1}{3}x^2 \ln x - 2x + 5$, $x > 0$. Show that it is convex for all $x \geq e^{-3/2}$.

11

## Relating Rates of Change 96

Find the rate of change of the area $A$ of a circle when $r = 3$ cm, given $dr/dt = 5$ cm s$^{-1}$.
First represent the statements mathematically.

"Rate of change of area"

Area rate

"Radius changes at a rate of 5"

Radius rate

"Area of a circle"

Area

**Tip**

When you see "rate", think differentiation with respect to time.

**Build the chain rule**

Copy the desired numerator and denominator into the diagonal positions, then fill the remaining positions with the intermediate variable $r$.

$$\frac{dA}{dt} = \frac{dA}{\square} \times \frac{\square}{dt}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

3 — Calculate

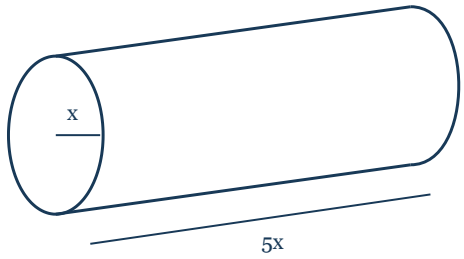
## Further Example — Expanding Cylinder 97

**Edexcel C4 June 2008, Q3**

A right circular cylindrical metal rod expands as it is heated. At time $t$ seconds its radius is $x$ cm and length $5x$ cm. Its cross-sectional area increases at $0.032\ \text{cm}^2\text{s}^{-1}$.

a. Find $dx/dt$ when $x = 2$, to 3 s.f. **(4 marks)**

b. Find the volume's rate of increase when $x = 2$. **(4 marks)**




It is common for a second part to connect a further pair of rates.

a

b

## Test Your Understanding — Expanding Cube 98

**June 2012, Q2**

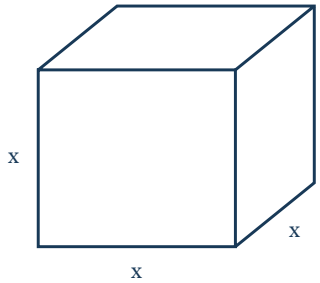
A metal cube expands uniformly as it is heated. At time $t$ seconds, each edge is $x$ cm and the volume is $V$ cm$^3$.

a. Show that $dV/dx = 3x^2$. **(1 mark)**

The volume increases at $0.048\ \text{cm}^3\text{s}^{-1}$.

b. Find $dx/dt$ when $x = 8$. **(2 marks)**

c. Find the surface area's rate of increase when $x = 8$. **(3 marks)**




a

b

c

**Tip**

$dx/dV = 1/(dV/dx)$.

**Note**

Setting up differential equations is covered in Integration. Leave Questions 5, 6, 8, 9, 12 and 14 until that topic has been studied.

1. Given $A = \frac{1}{4}\pi r^2$, $dr/dt = 6$, find $dA/dt$ when $r = 2$.

1

2. Given $y = xe^x$, $dx/dt = 5$, find $dy/dt$ when $x = 2$.

2



3. Given $r = 1 + 3\cos\theta$, $d\theta/dt = 3$, find $dr/dt$ when $\theta = \pi/6$.

3

4. Given $V = \frac{1}{3}\pi r^3$, $dV/dt = 8$, find $dr/dt$ when $r = 3$.

4

**STEP I 2009, Q5**

A right circular cone has radius $r$, height $h$, slant height $l$, volume $V$ and curved surface area $A$:

$$V = \frac{1}{3}\pi r^2 h, \qquad A = \pi r l.$$

i. With $A$ fixed, $r$ is chosen so that $V$ is stationary. Show that $A^2 = 3\pi^2 r^4$ and $l = \sqrt{3}r$.

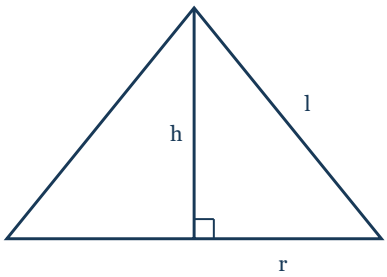
ii. Instead, $V$ is fixed and $r$ is chosen so that $A$ is stationary. Find $h$ in terms of $r$.

ii



**Solution to i**

With $A$ fixed, $l = A/(\pi r)$. A side view of the cone gives $l^2 = h^2 + r^2$.




Geometry

Stationary volume